SPL1 Project Proposal Form, 2025
Institute of Information Technology (IIT)
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Project Description:

The project aims to implement a **Recurrent Neural Network (RNN)** model from scratch in C++ to detect malicious **Domain Name System (DNS)** query behavior**.**

The system will**:**

- Collect and preprocess **DNS** query datasets containing both benign and malicious samples such as **Domain Generation Algorithm (DGA), Phishing, DNS Tunnelling and Command and Control (C2) communication.**
- Store and handle dataset information using CSV files for efficient reading and writing.
- Encode **DNS** queries into numerical or character-based sequences suitable for **RNN** processing**.**
- Implement core neural network components manually including
    - Activation functions
    - Training optimizer
    - Forward and backward propagation
- Train and test the model to classify **DNS** queries as benign or malicious**.**
- Design and implement the **Long Short Term Memory (LSTM)** architecture to handle long term dependencies in **DNS** sequences.
- Evaluate performance based on accuracy and error rate**.**

This project aims to understand and implement the mathematical and algorithmic concepts of **RNNs** to detect abnormal **DNS** queries independently of external machine learning libraries**.**

Languages or Tools to be used:

- C++ (for full implementation of the model)
- CSV data files (for dataset input and output)
- Basic file handling
- Custom built in math utilities

Supervisor's Name: Dr. Rezvi Shahariar

Signature of the supervisor: _____

Date: _____

Before the Midterm Presentation:

I confirm that the progress is satisfactory and I am forwarding it for midterm presentation.

Signature of the supervisor: _____

Date: _____

Midterm Presentation Feedback: